

Prayas JEE 3.0 2025

Maths

DPP: 2

Trigonometric Equation

Q1 If $r \sin \theta = 3, r = 4(1 + \sin \theta), 0 \leq \theta \leq 2\pi$,

then $\theta =$

- (A) $\frac{\pi}{6}, \frac{\pi}{3}$
 (B) $\frac{\pi}{6}, \frac{5\pi}{6}$
 (C) $\frac{\pi}{3}, \frac{\pi}{4}$
 (D) $\frac{\pi}{2}, \pi$

Q2 $\sin 6\theta + \sin 4\theta + \sin 2\theta = 0$, then $\theta =$

- (A) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{3}$
 (B) $\frac{n\pi}{4}$ or $n\pi \pm \frac{\pi}{6}$
 (C) $\frac{n\pi}{4}$ or $2n\pi \pm \frac{\pi}{6}$
 (D) None of these

Q3 If $\frac{1}{6}\sin \theta, \cos \theta$ and $\tan \theta$ are in G. P. then the general solution for θ is

- (A) $2n\pi \pm \frac{\pi}{3}$
 (B) $2n\pi \pm \frac{\pi}{6}$
 (C) $n\pi \pm \frac{\pi}{3}$
 (D) None of these

Q4 The general solution of $\sin x - \cos x = \sqrt{2}$ for any integer n is

- (A) $n\pi$
 (B) $2n\pi + \frac{3\pi}{4}$
 (C) $2n\pi$
 (D) $(2n + 1)\pi$

Q5 The set of value of x for which the expression

$$\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1 \text{ is}$$

- (A) ϕ
 (B) $\pi/4$
 (C) $(n\pi + \frac{\pi}{4}; n = 1, 2, 3 \dots)$

(D) $(2n\pi + \frac{\pi}{4}; n = 1, 2, 3 \dots)$

Q6 Total number of solutions of

$\cos x = \sqrt{1 - \sin 2x}$ in $[0, 2\pi]$, is equal to

- (A) 2
 (B) 3
 (C) 5
 (D) None of these

Q7 The number of solution of the equation

$\tan x + \sec x = 2 \cos x$ lying in the interval $(0, 2\pi)$ is.

- (A) 0
 (B) 1
 (C) 2
 (D) 3

Q8 The number of solutions of the equation

$|2 \sin x - \sqrt{3}|^2 \cos^2 x - 3 \cos x + 1 = 1$ in $[0, \pi]$ is

- (A) 2
 (B) 3
 (C) 4
 (D) 5

Q9 Equation $\sin^6 x + \cos^6 x = a^2$ has real solution, if

- (A) $a \in (-1, 1)$
 (B) $a \in (-1, \frac{-1}{2})$
 (C) $a \in (\frac{-1}{2}, \frac{1}{2})$
 (D) None of these

Q10 Find the general solutions of

$$2^{1+|\cos x|+|\cos x|^2+|\cos x|^3+\dots \text{ to } \infty} = 4$$

- (A) $n\pi \pm \frac{\pi}{6}$
 (B) $n\pi \pm \frac{\pi}{3}$
 (C) $(2n + 1)\frac{\pi}{2}$
 (D) $2n\pi - \frac{\pi}{3}$



Answer Key

Q1 (B)

Q2 (A)

Q3 (A)

Q4 (B)

Q5 (A)

Q6 (B)

Q7 (C)

Q8 (B)

Q9 (D)

Q10 (B)



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